

PRACTICAL 2

a)

Output :

```
manasvi@ManasviG:~$ nano prac2.c
manasvi@ManasviG:~$ gcc prac2.c
manasvi@ManasviG:~$ ./a.out
Source file descriptor: -1
Destination file descriptor: 3
File copied successfully..
manasvi@ManasviG:~$ |
```

Contents of source file :

```
manasvi@ManasviG:~$ cat a8
this is line 1
this is line2
this is line3
this is line 4
this is line 5
this is line 6
this is line 7
this is line 8
this is line 9
this is line 10
```

```
manasvi@ManasviG:~$ cat a20
this is line 1
this is line2
this is line3
this is line 4
this is line 5
this is line 6
this is line 7
this is line 8
this is line 9
this is line 10
```

b) Command

Strace

```
manasvi@ManasviG:~$ strace cat a8
execve("/usr/bin/cat", ["cat", "a8"], 0x7fffea846eef8 /* 25 vars */) = 0
brk(NULL)                               = 0x57d5e7ed3000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
0x70d8d4280000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27891, ...}) = 0
mmap(NULL, 27891, PROT_READ, MAP_PRIVATE, 3, 0) = 0x70d8d4279000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"...
, 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d8d4246
000
mmap(0x70d8d424d000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x7000) = 0x70d8d424d000
mmap(0x70d8d426e000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x28000) = 0x70d8d426e000
mmap(0x70d8d4275000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D
ENYWRITE, 3, 0x2f000) = 0x70d8d4275000
mmap(0x70d8d4277000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_A
NONYMOUS, -1, 0) = 0x70d8d4277000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEX
EC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"...
, 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d8d4218
000
```

```
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0\0@\0\0\0\0\0\0\0" ..
., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d8d3e0
0000
mmap(0x70d8d3e28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP
_DENYWRITE, 3, 0x28000) = 0x70d8d3e28000
mmap(0x70d8d3fc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE
, 3, 0x1c0000) = 0x70d8d3fc0000
mmap(0x70d8d400e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x20d000) = 0x70d8d400e000
mmap(0x70d8d4014000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_
ANONYMOUS, -1, 0) = 0x70d8d4014000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcr2-8.so.0", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0" ...,
, 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d8d4046
000
mmap(0x70d8d4049000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x3000) = 0x70d8d4049000
mmap(0x70d8d40c5000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
, 3, 0x7f000) = 0x70d8d40c5000
mmap(0x70d8d40f0000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D
ENYWRITE, 3, 0xa9000) = 0x70d8d40f0000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x70d8d4044000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x70d8d4041000
arch_prctl(ARCH_SET_FS, 0x70d8d4041b00) = 0
set_tid_address(0x70d8d4042128) = 1357
set_robust_list(0x70d8d4041de0, 24) = 0
rseq(0x70d8d40416a0, 0x20, 0, 0x53053053) = 0
mprotect(0x70d8d400e000, 16384, PROT_READ) = 0
mprotect(0x70d8d40f0000, 4096, PROT_READ) = 0
mprotect(0x70d8d4216000, 4096, PROT_READ) = 0
mprotect(0x70d8d4244000, 4096, PROT_READ) = 0
mprotect(0x70d8d4275000, 4096, PROT_READ) = 0
mprotect(0x57d5c30c4000, 1462272, PROT_READ) = 0
mprotect(0x70d8d42c5000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINI
TY}) = 0
getrandom("\x45\x95\x6b\x69\x5c\xf5\xb0\x2f", 8, GRND_NONBLOCK) = 8
```

```

mprotect(0x70d8d4273000, 4096, PROT_READ) = 0
mprotect(0x57d5c30c4000, 1462272, PROT_READ) = 0
mprotect(0x70d8d42c5000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\x45\x95\x6b\x69\x5c\xf5\xb0\x2f", 8, GRND_NONBLOCK) = 8
munmap(0x70d8d4279000, 27891) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0xa6c10547, 0xb517db5b]}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NODEV|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7fff208b94a0) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x57d5e7ed3000
brk(0x57d5e7ef4000) = 0x57d5e7ef4000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"... , 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-"... , 1024) = 1024
read(3, "e 0 0\ndepts /dev/pts devpts rw,"... , 1024) = 1024
read(3, "osuid,nodev,noexec,relatime,nosy"... , 1024) = 1024
read(3, "n/user/1000 tmpfs rw,nosuid,node"... , 1024) = 104
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Time out)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x70d8d3e45cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "57d5e375d000-57d5e384d000 r-x 0"..., 1024) = 1024

```